

J McCann & Co Ltd · KTP Associate Task

Deliverable 5

DevOps Pipeline

Commit to production in twelve minutes — every change linted, tested, built and traceable, with a clear path from a developer's keyboard to the operative's phone.

PROJECT	Backpack Prototype
DOCUMENT	5 — DevOps Pipeline
VERSION	1.0
PREPARED FOR	Dr N. Caldwell, University of Suffolk

1. Approach

Backpack uses a trunk-based workflow with short-lived feature branches and a protected main branch. The aim is simple: every change is automatically linted, tested, built and traceable before it can reach an operative in the field.

2. Continuous Integration

Opening a pull request triggers a GitHub Actions workflow. It checks out the code, runs flutter analyze with zero warnings permitted, executes the unit and widget tests — including the cost-engine tests — and scans dependencies for known vulnerabilities. It then builds the web and mobile artifacts. A branch cannot merge until all checks pass and a code owner has reviewed it, so regressions are caught before they spread.

3. Continuous Delivery

Merges promote the build through three environments: a per-pull-request dev preview, a staging site where the QS team can smoke-test against realistic mock data, and production. Promotion to production sits behind a manual approval gate, then deploys automatically to Firebase Hosting on a tagged release using semantic versioning.

4. Observability

Once live, Crashlytics, Performance Monitoring and Cloud Logging report back continuously, with alerts routed to the developer so issues surface within minutes rather than days. Auto-rollback is triggered if service-level objectives are breached, closing the loop from production back to the next fix.

5. Why this fits J McCann

Operatives depend on the app on-site, often with poor connectivity, so reliability matters more than raw release speed. The automated gates stop a broken build reaching a live job, and the staging environment lets the QS team validate cost-affecting changes safely before they go anywhere near real work. The targets below keep the pipeline honest.

DORA targets

METRIC	TARGET
Lead time for changes	Under 1 day
Deployment frequency	Daily
Mean time to restore (MTTR)	Under 1 hour
Change failure rate	Under 10%

End of document.